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물질안전보건자료
(Material Safety Data Sheet)

Product Name	Perfume base

1. Information about chemical products and companies

A. Product name: Perfume base

B. Recommended uses and restrictions on use of the product

Recommended Uses of the Product	No data
Restrictions on use of the product	No data

D. Supplier information (in case of imported products, please provide information on domestic suppliers who can be contacted in an emergency)

Company name	No data
address	No data
Emergency phone number	No data

2. Harmfulness/Danger

A. Classification of hazards and risks

Flammable liquids: Category 2

Serious eye damage/eye irritation: Category 2

Carcinogenicity: Category 1A

B. Warning label items including precautionary statements

Pictograph



Signal word: danger

Hazardous/Dangerous Statements:

H225 Highly flammable liquid and vapour

H319 Causes serious eye irritation

H350 May cause cancer

Precautionary statement

prevention:

P201 Obtain special instructions before use.

P202 Do not handle until all safety precautions have been read and understood.

P210 Keep away from heat/sparks/open flames/hot surfaces - No smoking

P233 Keep container tightly closed.

P240 Ground or bond container and receiving equipment.

P241 Use explosion-proof electrical, ventilation, lighting, (...) equipment.

P242 Use only non-sparking tools.

P243 Take precautions against static discharge.

P264 Wash affected area thoroughly after handling.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

react:

P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Wash skin with water/shower.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P308+P313 IF exposed or concerned: Get medical advice/attention.

P337+P313 If eye irritation persists: Get medical advice/attention.

P370+P378 In case of fire: Use (…) to extinguish.

save:

P403+P235 Store in a well-ventilated place. Keep cool.

P405 Store in a locked storage area.

dispose:

P501 Dispose of contents/container in accordance with applicable local/national regulations.

Ethanol

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Health	1
fire	3
Reactivity	No data

Propylene glycol

Health	0
fire	1
Reactivity	0

WATER

Health	0
fire	0
Reactivity	0

3. Name and content of ingredients

Material name	Alias (common name)	CAS number	content(%)
Ethanol	Ethyl alcohol Ethyl alcohol	64-17-5	80
Propylene glycol	1,2-PROPANEDIOL;	57-55-6	15
WATER	DIHYDROGEN OXIDE;	7732-18-5	5

4. Emergency measures

A. When it gets into your eyes	If in eyes, rinse cautiously with water for several minutes. Remove contact lenses, if possible. Wash your stomach. If eye irritation persists, seek medical advice/attention.
B. When in contact with skin	If on skin (or hair): Take off all contaminated clothing. Wash skin with water. Take a shower. Get emergency medical attention. Remove contaminated clothing and shoes and isolate the contaminated area. Prevent spread of contamination in case of minor skin contact In case of burns, cool the affected area with cold water for as long as possible and remove any clothing stuck to the skin. Don't do it Wash your skin with soap and water.
When inhaled	If exposed or concerned, seek medical advice/attention. Move to a place with fresh air. Keep me warm and comfortable
A. When I ate it	If exposed or concerned, seek medical advice/attention. If the substance has been ingested or inhaled, do not perform mouth-to-mouth artificial respiration, but provide proper breathing. Use medical equipment
B. Other doctor's notes	In case of exposure, contact medical staff and take special emergency measures such as follow-up investigation. Ensure that medical personnel are aware of the substance and take protective measures.

5. How to respond in case of explosion or fire

A. Appropriate (inappropriate) extinguishing agent	
Appropriate (inappropriate) digestive aids	When extinguishing fires involving this substance, use alcohol foam, carbon dioxide or water spray. When extinguishing by suffocation, use dry sand or soil.
B. Specific hazards arising from chemicals	
Specific hazards arising from chemicals	Highly flammable liquids and vapors May polymerize violently, causing fire and explosion. Vapors can be transferred to an ignition source and ignite. During burning, irritating and highly toxic gases may be generated by thermal decomposition or combustion. Can form explosive mixtures at or above the flash point Container may explode when heated. Highly flammable: Easily ignited by heat, sparks, or flames.

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Specific hazards arising from chemicals	Leakage may pose a fire/explosion hazard.
	Risk of steam explosion indoors, outdoors and in sewers
	Some can burn but do not ignite easily
	Vapors can form explosive mixtures with air.
	Non-flammable, the substance itself does not burn, but may decompose when heated to produce corrosive/toxic fumes.
D. Protective equipment and precautions to wear when fighting fires	
Ethanol	Rescuers should wear appropriate protective equipment.
	Extinguish the fire from a safe distance away from the area.
	Be careful, most are lighter than water.
	Most vapors are heavier than air, so they will spread along the ground and may accumulate in low-lying areas or confined spaces.
	If it is safe to do so, move containers from the fire area.
	Do not pour water directly onto exposed surfaces or safety devices as this may cause freezing in the event of a tank fire.
	In case of tank fire, extinguish from maximum distance or use unmanned fire extinguishing equipment.
	In case of a tank fire, cool the container with plenty of water even after the fire has been extinguished.
	If there is a loud noise from the pressure relief device or the tank becomes discolored in the event of a tank fire, vacate immediately.
	In case of a tank fire, stay away from the burning tank.
	In the case of a large-scale tank fire, use unmanned fire extinguishing equipment, and if that is not possible, stand back and let the fire burn.
Propylene glycol	Rescuers should wear appropriate protective equipment.
	Extinguish the fire from a safe distance away from the area.
	Be careful, it may be transported melted.
	Dig a trench to contain the digester and prevent the material from scattering.
	If it is safe to do so, move containers from the fire area.
	In case of tank fire, extinguish from maximum distance or use unmanned fire extinguishing equipment.
	In case of a tank fire, cool the container with plenty of water even after the fire has been extinguished.
	If there is a loud noise from the pressure relief device or the tank becomes discolored in the event of a tank fire, vacate immediately.
	In case of a tank fire, stay away from the burning tank.
	In the case of a large-scale tank fire, use unmanned fire extinguishing equipment, and if that is not possible, stand back and let the fire burn.
WATER	In case of a tank fire, cool the container with plenty of water even after the fire has been extinguished.
	If there is a loud noise from the pressure relief device or the tank becomes discolored in the event of a tank fire, vacate immediately.
	In case of a tank fire, stay away from the burning tank.
	If the container heats up and explodes, the sprayed water can cause burns to skin and eyes.

6. How to respond in case of a leak

a. Precautions and protective equipment required to protect the human body	Very fine particles can cause fire or explosion, so eliminate all sources of ignition.
	Wipe up spills immediately and follow safety precautions as directed.
	Isolate the contaminated area.
	Anyone who does not need to enter or is not wearing protective equipment is prohibited from entering.
	Eliminate all sources of ignition.
	Always ground all equipment when handling materials.
	If it is not dangerous, stop the leak.
	Do not touch damaged containers or spilled material without wearing appropriate protective clothing.
	Vapor suppressing foam can be used to reduce vapor generation.
	Cover with plastic sheet to prevent spreading.
	Be aware of substances and conditions to avoid.
B. Measures necessary to protect the environment	Prevent entry into waterways, sewers, basements, and confined spaces.
D. Method of purification or removal	Build a levee and collect water for digestion.

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D. Method of purification or removal	Absorb spills with inert material (e.g. dry sand or earth) and use a chemical waste disposal facility. Put it in the oven. Absorb liquid and wash contaminated area with detergent and water. In case of a large spill, create a trench away from the liquid spill. Use clean, explosion-proof tools to collect the absorbed material.

7. Handling and storage methods

A. Safety handling instructions	Do not handle until you have read and understood all safety precautions. Use explosion-proof electrical, ventilation, lighting, (...) equipment. Use only non-sparking tools. Take precautions against static electricity. Wash the affected area thoroughly after handling. Do not apply pressure, cut, weld, solder, join, drill, grind or expose to heat, flame, sparks, Do not expose to static electricity or other ignition sources. All MSDS/label precautions must be followed as product residue may remain after the container is emptied. Follow the teeth. Use with care when handling/storing. Open the cap carefully before opening. Do not breathe vapors from heated material. Do not enter storage areas without adequate ventilation. Always ground all equipment when handling materials. Be aware of substances and conditions to avoid. Be careful of the heat There is a risk of oxygen deficiency when working in confined spaces at low altitudes, so measure the oxygen concentration in the air while working. Clean and ventilate
B. Safe storage method	Keep away from heat, sparks, flames and hot surfaces - No smoking Keep container tightly closed. Store in a well-ventilated place and keep at low temperature. Store in a locked storage location. Empty drums should be completely drained, properly sealed and immediately returned to the drum regulator or properly Place it.

8. Exposure prevention and personal protective equipment

A. Exposure standards for chemical substances, biological exposure standards, etc.

Domestic regulations

Ethanol	TWA - 1000ppm
Propylene glycol	No data
WATER	No data

ACGIH regulations

Ethanol	STEL 1000 ppm
Propylene glycol	No data
WATER	No data

Biological exposure standards

Ethanol	No data
Propylene glycol	No data
WATER	Not applicable

Other exposure criteria

Ethanol	No data
Propylene glycol	No data
WATER	No data

b. Appropriate engineering management	Use process isolation, local exhaust, or other engineering methods to control air levels below exposure limits. Take care of it.

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b. Appropriate engineering management	Facilities storing or using this substance must be equipped with eyewash facilities and safety showers.
D. Personal protective equipment	
Respiratory Protection	
Ethanol	A product certified by the Korea Occupational Safety and Health Agency that matches the physical and chemical properties of the gas/liquid to which it is exposed. Wear breathing protection.
Ethanol	If the exposure concentration is lower than 10,000 ppm, use a half-size respirator equipped with an appropriate filter or purifier. Wear protective gear.
Ethanol	Non-disposable type with appropriate filter or purifier if exposure concentration is lower than 25,000 ppm Wear a (loose-fitting) hooded/helmet-type powered respirator or a continuous-flow dust mask. Do it
Ethanol	If the exposure concentration is lower than 50,000 ppm, a full-face or Wear a powered half-circuit or air-supplied continuous-flow/pressure-demand half-circuit respirator. Sio
Ethanol	If the exposure concentration is lower than 1,000,000 ppm, use a full-face or full-face type respirator equipped with an appropriate filter or purifier. Wear a helmet/hood type, pressure-demand type respirator.
Ethanol	If the exposure concentration is lower than 10000000 ppm, self-processing with an appropriate filter or purification tank Wear self-contained breathing apparatus (SCBA) or pressure-demand self-contained breathing apparatus (SCBA).
Propylene glycol	A respirator certified by the Occupational Safety and Health Agency that matches the physicochemical properties of the gas/liquid to which it is exposed. Wear protective gear
Propylene glycol	For gaseous/liquid substances, the following respiratory protection is recommended: - Isolation type full-face gas mask (for organic compounds (for acid gases in case of acid gases)) or isolation type Half-face gas mask (for organic compounds (for acid gases)) or direct-connected full-face mask Gas mask (for organic compounds (for acid gases)) or half-face gas mask (for organic compounds) For gaseous compounds (for acidic gases) or electric gas masks
Propylene glycol	In case of oxygen deficiency (<19.5%), wear a supplied-air mask or self-contained breathing apparatus.
WATER	A respirator certified by the Occupational Safety and Health Agency that matches the physicochemical properties of the gas/liquid to which it is exposed. Wear protective gear
WATER	For gaseous/liquid substances, the following respiratory protection is recommended: - Isolation type full-face gas mask (for organic compounds (for acid gases in case of acid gases)) or isolation type Half-face gas mask (for organic compounds (for acid gases)) or direct-connected full-face mask Gas mask (for organic compounds (for acid gases)) or half-face gas mask (for organic compounds) For gaseous compounds (for acidic gases) or electric gas masks
WATER	In case of oxygen deficiency (<19.5%), wear a supplied-air mask or self-contained breathing apparatus.

9. Physical and chemical properties

A. Appearance	
Constellation	No data
color	No data
B. Smell	No data
D. Odor threshold	No data
La. pH	No data
Ma. Melting point/freezing point	No data
B. Initial boiling point and boiling range	No data
4. Flashpoint	No data
Ah. Evaporation rate	No data
Yes. Flammability (solid, gas)	No data
Tea. Upper/lower limits of ignition or explosion range	No data
Car. Vapor pressure	No data
t. Solubility	No data
Wave. Vapor density	No data
Ha. Specific gravity	No data
. n-octanol/water partition coefficient	No data
You. Autoignition temperature	No data
More. Decomposition temperature	No data
Ru. Viscosity	No data
What is molecular weight?	No data

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Ethanol

A. Appearance	
Constellation	liquid
color	Colorless
B. Smell	smell of wine or whiskey
D. Odor threshold	10 ppm
La. pH	No data
Ma. Melting point/freezing point	-114.1 ÿ
B. Initial boiling point and boiling range	78.5 ÿ
4. Flashpoint	13 ÿ (cc)
Ah. Evaporation rate	No data
Yes. Flammability (solid, gas)	No data
Tea. Upper/lower limits of ignition or explosion range	3.3 / 19 %
Car. Vapor pressure	59.3 mmHg (25ÿ)
t. Solubility	100 g/100ml (25ÿ)
Wave. Vapor density	1.59 (air=1)
Ha. Specific gravity	0.7893 (20ÿ)
. n-octanol/water partition coefficient	-0.31
You. Autoignition temperature	363 ÿ
More. Decomposition temperature	(enthalpy change 1336.8 kJ/mol at 25ÿ)
Ru. Viscosity	1.074 (20ÿ, mPa s)
What is molecular weight?	46.07

Propylene glycol

A. Appearance	
Constellation	Liquid (hygroscopic)
color	Colorless
B. Smell	Odorless
D. Odor threshold	No data
La. pH	No data
Ma. Melting point/freezing point	-59 ÿ
B. Initial boiling point and boiling range	188.2 ÿ
4. Flashpoint	99 ÿ (cc, 1075 ÿ oc)
Ah. Evaporation rate	No data
Yes. Flammability (solid, gas)	combustibility
Tea. Upper/lower limits of ignition or explosion range	12.6 / 2.6 %
Car. Vapor pressure	0.129 mmHg (at 25 ÿ)
t. Solubility	1000000 ÿ/ÿ (at 20 ÿ)
Wave. Vapor density	2.6
Ha. Specific gravity	1.0361
. n-octanol/water partition coefficient	-0.92
You. Autoignition temperature	371ÿ
More. Decomposition temperature	403 ÿ
Ru. Viscosity	0.581 cP (at 20 ÿ)
What is molecular weight?	76.09

WATER

A. Appearance	
Constellation	liquid
color	Colorless (transparent)
B. Smell	Odorless
D. Odor threshold	(not applicable)

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La. pH	7
Ma. Melting point/freezing point	0 ÿ
B. Initial boiling point and boiling range	100ÿ
4. Flashpoint	(not applicable)
Ah. Evaporation rate	No data
Yes. Flammability (solid, gas)	Not applicable
Tea. Upper/lower limits of ignition or explosion range	- / - (not applicable)
Car. Vapor pressure	23.8 mmHg (25ÿ)
t. Solubility	100 g/100ml
Wave. Vapor density	No data
Ha. Specific gravity	1
. n-octanol/water partition coefficient	-1.38
You. Autoignition temperature	No data
More. Decomposition temperature	No data
Ru. Viscosity	No data
What is molecular weight?	18.02

10. Stability and Responsiveness

A. Chemical stability and possibility of hazardous reactions

Ethanol	Highly flammable liquids and vapors
Ethanol	May polymerize violently, causing fire and explosion.
Ethanol	Can form explosive mixtures at or above the flash point
Ethanol	Container may explode when heated.
Ethanol	Highly flammable: Easily ignited by heat, sparks, or flames.
Ethanol	Leakage may pose a fire/explosion hazard.
Ethanol	Risk of steam explosion indoors, outdoors and in sewers
Ethanol	Vapors can form explosive mixtures with air.
Ethanol	Vapors can travel to an ignition source and flash back.
Ethanol	Steam can cause dizziness or suffocation without awareness.
Ethanol	Irritating to skin and eyes or causing burns when inhaled or in contact
Propylene glycol	Container may explode when heated.
Propylene glycol	Some can burn but do not ignite easily
Propylene glycol	Non-flammable, the substance itself does not burn, but may decompose when heated to produce corrosive/toxic fumes. hmm
Propylene glycol	May produce irritating, corrosive and toxic gases in case of fire
WATER	Stable under normal temperature and pressure conditions
WATER	Container may explode when heated.

B. Conditions to avoid

Ethanol	Keep away from heat, sparks, flames and hot surfaces - No smoking
Propylene glycol	Ignition sources such as heat, sparks, or flames
WATER	heat, pollution

D. Substances to avoid

Ethanol	No data
Propylene glycol	Combustible substances, reducing substances
WATER	water reactive substance

A. Harmful substances produced during decomposition

Ethanol	During burning, irritating and highly toxic gases may be generated by thermal decomposition or combustion.
Propylene glycol	During burning, irritating and highly toxic gases may be generated by thermal decomposition or combustion.
Propylene glycol	Corrosive/toxic fumes

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WATER	No data

11. Information on toxicity

a. Information on likely routes of exposure;

Ethanol	No data
Propylene glycol	No data
WATER	No data

B. Health Hazard Information

Acute toxicity

oral-

Ethanol	LD50 7060 ÿ/ÿ Rat (OECD Guideline 401)
Propylene glycol	LD50 22000 mg/kg Rat
WATER	LD50 90000 ÿ/kg Rat (LD50 > 90 ml/kg (Rat))

Transdermal

Ethanol	No data
Propylene glycol	LD50 > 16000 ÿ/ÿ Rabbit
WATER	No data

inhalation

Ethanol	Vapor LC50 30300 mg/ÿ 4 hr Mouse (OECD Guideline 403)
Propylene glycol	No data
WATER	No data

Skin corrosion or irritation

Ethanol	Skin corrosion/irritation test using rabbit showed no irritation (OECE) Guideline 404, GLP)
Propylene glycol	Skin irritation test results (rabbit) Non-irritating
WATER	Not applicable

Severe eye damage or irritation

Ethanol	Results of severe eye damage/irritation tests using rabbits showed conjunctivitis, conjunctival edema, iris damage, and corneal damage. Occurred (Conjunctival index: 2.1, Iris index: 0.44, Conjunctival edema index: 1.3, Corneal index: 1.1, OECD Guideline 405)
Propylene glycol	Eye irritation test results (rabbit) Very weak irritation
WATER	Not applicable

Respiratory hypersensitivity

Ethanol	No data
Propylene glycol	No data
WATER	Not applicable

Skin sensitivity

Ethanol	Skin sensitization test results using mice (male/female) showed that no skin sensitization occurred.
Propylene glycol	Human/Draize Test: No hypersensitivity
WATER	Not applicable

Carcinogenic

Industrial Safety and Health Act

Ethanol	No data
Propylene glycol	No data
WATER	No data

Ministry of Employment and Labor Notice

Ethanol	1A ((Limited to alcohol consumption))
Propylene glycol	No data
WATER	No data

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IARC

Ethanol	1 (Ethanol in alcoholic beverages)
Propylene glycol	No data
WATER	No data

OSHA

Ethanol	No data
Propylene glycol	No data
WATER	No data

ACGIH

Ethanol	A3
Propylene glycol	No data
WATER	No data

NTP

Ethanol	No data
Propylene glycol	No data
WATER	No data

EU CLP

Ethanol	No data
Propylene glycol	No data
WATER	No data

Germ cell mutagenicity

Ethanol	Positive results from in vivo rodent lethality test (OECD Guideline 478) Negative results from spot test using mice in vivo (OECD Guideline 484) Negative result of micronucleus test using mammalian erythrocytes in vivo (OECD Guideline 474) Negative results of chromosome aberration test using in vivo mammalian bone marrow cells (OECD Guideline 475)
Propylene glycol	In vitro - Salmonella typhimurium/TA 98, TA100, TA1535, TA1537 (revertant mutation This test; Ames test): Negative, Human/sister chromatid exchange test: Negative
	castle)
WATER	Not applicable

Reproductive toxicity

Ethanol	No significant effects were observed in the developmental toxicity/teratogenicity/maternal toxicity test using rats (male) (developmental toxicity Sex NOAEL = 4000 mg/kg, Teratogenic NOAEL = 5200 mg/kg, Teratogenic LOAEL = 8200mg/kg)(OECD Guideline 415)
Propylene glycol	When administered to pregnant rabbits at 1230 mg/kg in the diet for 10 days, there was no effect on the fertility rate or the fetus. There was no effect on maternal survival rate. Fetal developmental toxicity without maternal toxicity is most significantly skeletal and teratogenic, and in mice ÿ500 mg/kg/day, rats ÿ1,000 mg/kg/day. Fetal weight and survival rate Effects occur at higher concentrations.
WATER	Not applicable

Specific target organ toxicity (single exposure)

Ethanol	Oral toxicity test results using rabbits showed that eye twitching and vestibular function were suppressed.
Propylene glycol	Non-toxic syndrome is central nervous system depression during anesthesia. There is no target organ.
WATER	Not applicable

Specific target organ toxicity (repeated exposure)

Ethanol	Repeated oral toxicity test (98d) using rats (male/female) showed no significant effects
Propylene glycol	When exposed to rats for 90 days, body weight and food intake decreased, but clinical-chemical and hematological No changes in numbers. Organs (liver, kidneys, pancreas, lungs) and blood have no toxicological effects.
WATER	Not applicable

Aspiration hazard

Ethanol	No data
Propylene glycol	No data
WATER	Not applicable

Other harmful effects

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Ethanol	No data
Propylene glycol	No data
WATER	No data

12. Impact on the environment

A. Ecotoxicity

Pisces

Ethanol	LC50 0.0142 ÿ/ÿ 96 hr Pimephales promelas (US EPA method E03-05)
Propylene glycol	LC50 710 ÿ/ÿ 96 hr Oncorhynchus mykiss
WATER	No data

shellfish

Ethanol	LC50 5012 ÿ/ÿ 48 hr Ceriodaphnia dubia (other guideline: ASTM E729-80)
Propylene glycol	EC50 > 1000 ÿ/ÿ 48 hr Daphnia magna
WATER	No data

birds

Ethanol	ErC50 275 ÿ/ÿ 72 hr Chlorella vulgaris (OECD Guideline 201)
Propylene glycol	EC50 > 1000 ÿ/ÿ 72 hr Selenastrum capricornutum
WATER	No data

B. Persistence and degradability

Residuality

Ethanol	log Kow -0.35
Propylene glycol	log Kow -0.92
WATER	log Kow -1.38

Decomposability

Ethanol	No data
Propylene glycol	No data
WATER	No data

D. Bioaccumulation

Concentration

Ethanol	BCF 1
Propylene glycol	BCF<1
WATER	No data

Biodegradable

Ethanol	71% (biodegradable)
Propylene glycol	> 60 (%) 10 days
WATER	No data

A. Soil mobility

Ethanol	No data
Propylene glycol	No data
WATER	No data

B. Other adverse effects

Ethanol	Crustaceans: Daphnia magna: NOEC, 9d, = 9.6 mg/L Bird: Skeletonema costatum: NOEC, 120h, = 3240mg/L
Propylene glycol	No data
WATER	No data

13. Precautions when disposing

A. Disposal method

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Ethanol	Handle it in one of the following ways: 1. Incinerate. 2. After processing by evaporation/concentration method, incinerate the residue. 3. After purifying by separation, distillation, extraction, and filtration, incinerate the residue. 4. Treat using reactions of neutralization, oxidation, reduction, polymerization, and condensation. 5. The residue is incinerated or reprocessed through coagulation, sedimentation, filtration, or dehydration. Incinerate the water.
Propylene glycol	Dispose of contents and containers in accordance with regulations specified in the Waste Management Act.
WATER	Dispose of contents and containers in accordance with regulations specified in the Waste Management Act.

B. Precautions for disposal

Ethanol	Dispose of contents container (in accordance with relevant laws and regulations).
Propylene glycol	Dispose of contents container (in accordance with relevant laws and regulations).
WATER	Dispose of contents container (in accordance with relevant laws and regulations).

14. Information required for transportation

A. UN Number

Ethanol	1170
Propylene glycol	No UN transport hazardous materials classification information available
WATER	No UN transport hazardous materials classification information available

B. Proper shipping name

Ethanol	ETHANOL (ETHYL ALCOHOL) or ETHANOL SOLUTION (ETHYL ALCOHOL SOLUTION))
Propylene glycol	Not applicable
WATER	Not applicable

D. Risk ratings in transport

Ethanol	3
Propylene glycol	Not applicable
WATER	Not applicable

A. Courage level

Ethanol	ÿ
Propylene glycol	Not applicable
WATER	Not applicable

B. Marine pollutants

Ethanol	Not applicable
Propylene glycol	No data
WATER	No data

B. Special safety precautions that users need to know or require in relation to transportation or means of transportation.

Emergency measures in case of fire

Ethanol	FE
Propylene glycol	Not applicable
WATER	Not applicable

Emergency measures in case of leakage

Ethanol	SD
Propylene glycol	Not applicable
WATER	Not applicable

15. Legal Regulation Status

A. Regulations under the Industrial Safety and Health Act

Ethanol	Substances subject to submission of Process Safety Report (PSM)
Ethanol	Exposure standard setting substance
Propylene glycol	No data
WATER	No data

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B. Regulations under the Chemical Substances Control Act

Ethanol	No data
Propylene glycol	No data
WATER	No data

D. Regulations under the Hazardous Materials Safety Management Act

Ethanol	400L of Class 4 alcohol
Propylene glycol	4000ÿ of Class 4, Third Class Petroleum (water-soluble liquid)
WATER	No data

A. Regulations under the Waste Management Act

Ethanol	Designated waste
Propylene glycol	No data
WATER	No data

B. Regulations under other domestic and foreign laws

Domestic regulations

Persistent Organic Pollutants Management Act

Ethanol	Not applicable
Propylene glycol	Not applicable
WATER	Not applicable

Foreign regulations

US Management Information (OSHA Regulations)

Ethanol	Not applicable
Propylene glycol	Not applicable
WATER	Not applicable

US Management Information (CERCLA Regulations)

Ethanol	Not applicable
Propylene glycol	Not applicable
WATER	Not applicable

US Management Information (EPCRA 302 Regulation)

Ethanol	Not applicable
Propylene glycol	Not applicable
WATER	Not applicable

US Management Information (EPCRA 304 Regulation)

Ethanol	Not applicable
Propylene glycol	Not applicable
WATER	Not applicable

US Management Information (EPCRA 313 Regulation)

Ethanol	Not applicable
Propylene glycol	Not applicable
WATER	Not applicable

US Management Information (Rotterdam Convention Substances)

Ethanol	Not applicable
Propylene glycol	Not applicable
WATER	Not applicable

US Management Information (Stockholm Convention Substances)

Ethanol	Not applicable
Propylene glycol	Not applicable
WATER	Not applicable

US Management Information (Montreal Protocol Substances)

Ethanol	Not applicable
Propylene glycol	Not applicable

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WATER	Not applicable

EU classification information (confirmed classification results)

Ethanol	Flam. Liq. 2
Propylene glycol	Not applicable
WATER	Not applicable

EU classification information (risk phrases)

Ethanol	H225
Propylene glycol	Not applicable
WATER	Not applicable

EU classification information (safety phrases)

Ethanol	Not applicable
Propylene glycol	Not applicable
WATER	Not applicable

16. Other Notes

A. Source of data

Ethanol

HSDB(Constellation)

HSDB(color)

HSDB(B. Smell)

HSDB(Ma. Melting point/freezing point)

HSDB (Ba. Initial boiling point and boiling point range)

HSDB (Incident Point)

IPCS(Inflammability and Explosion Limits)

HSDB(Car. Vapor Pressure)

HSDB(Other Solubility)

HSDB(Pa. Vapor Density)

HSDB(lower)

HSDB(n-octanol/water partition coefficient)

IPCS(Initial Ignition Temperature)

HSDB(Decomposition temperature)

HSDB(Russian viscosity)

HSDB(Molecular weight)

ECHA (oral)

ECHA (inhalation)

ECHA (skin corrosion or irritation)

ECHA (serious eye damage or irritation)

ECHA (Skin Sensitization)

ECHA (germ cell mutagenicity)

ECHA (Reproductive Toxicity)

HSDB (Specific target organ toxicity (single exposure))

ECHA (Specific target organ toxicity (repeated exposure))

ECHA (fish)

ECHA (crustaceans)

ECHA (Algae)

ECHA (residual)

ECHA (concentrated)

ECHA (MA. Other adverse effects)

Propylene glycol

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ICSC(Constellation)

ICSC(color)

ICSC (B. Smell)

ICSC(Melting point/freezing point)

ICSC(Initial boiling point and boiling range)

ICSC (Incident Point)

ICSC(Inflammability (solid, gas))

ICSC(Inflammability or Explosion Limit Upper/Lower)

ChemIDplus(Car. Vapor Pressure)

ChemIDplus (Other Solubility)

ICSC(Pa. Vapor Density)

ICSC(lower, specific gravity)

ICSC(n-octanol/water partition coefficient)

ICSC(Instant Ignition Temperature)

The Chemical Database, The Department of Chemistry at the University of Akron (http://ull.chemistry.uakron.edu/erd) (decomposition temperature)

HSDB(Russian viscosity)

ICSC(Molecular weight)

SIDS(oral)

International Uniform Chemical Information Database (IUCLID) (http://ecb.jrc.it/esis) (transdermal)

SIDS (skin corrosion or irritation)

SIDS (Severe Eye Damage or Irritation)

Intermational Program on Chemical Safety (IPCS INCHEM) (http://www.inchem.org/) (skin sensitization)

National Library of Medicine/Chemical Carcinogenesis Research Information System (NLM/CCRIS) (http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?CCRIS)(germ cell mutagenicity)

National Library of Medicine/genetic toxicology(NLM/GENETOX)(http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?GENETOX) (Germ cell mutagenicity)

National Library of Medicine/Agency for Toxic Substances and Disease Registry (NLM/ATSDR) (http://www.atsdr.cdc.gov /MHMI/mmg111.html) (reproductive toxicity)

International Uniform ChemicaL Information Database(IUCLID)(http://ecb.jrc.it/esis)(Specific target organ toxicity (single exposure))

International Uniform ChemicaL Information Database(IUCLID)(http://ecb.jrc.it/esis)(Specific target organ toxicity (repeated exposure))

ECOTOX (Fish)

ECOTOX (crustaceans)

National Institute of Technology and Evaluation (NITE) (http://www.safe.nite.go.jp/ghs/h18_bunrui.html) (birds)

ICSC(residual)

SIDS(Concentrated)

SIDS(Biodegradable)

WATER

NLM

B. Date of first creation	2018-05-01
D. Number of revisions and date of last revision	
Number of revisions	episode
Last revised date	0

Ra. Other

• The written Material Safety Data Sheet (MSDS) was edited and partially modified with reference to the MSDS provided by the Korea Occupational Safety and Health Agency.
This is the data.